TrueRehab-RF: what to do first

Priorities ranked in two groups: A = promised in the proposal but not done, B = problems in the current hardware and software. Status as of 2026-11-09.

1 **First full recording session**
Radar + MyoWare + IMU/webcam on a person. Unlocks A1–A3 and A5.

2 **Run the Pi**
Flash SD, install, self-test, live radar. Needed for the demo (A4).

3 **Decide phase output**
Can the radar give phase or micro-vibration? Decides A2 and A10.

4 **Fix or drop the IMU**
Wiring fails on the arm. Do not let it block step 1.

5 **Ethics before recruiting**
Consent for healthy volunteers. Needed before more than the team is recorded.

A. Promised in the proposal, not done

Ranked by how much the demo and the claim depend on it

1. **Phase 1: elbow-flexion proof of concept** — partly
   Radar sees curls, but no comparison against IMU or video ground truth yet.
2. **Phase 2: muscle signal investigation** — not done
   Radar vs sEMG across reps to failure. MyoWare not yet worn.
3. **Phase 3: compensation detection** — not done
   Shoulder elevation and trunk lean, labelled by pose or IMU.
4. **Phase 7: demonstration system on the Pi** — partly
   Dashboard, bridge and replay exist. Not run on the Pi or live radar.
5. **Phase 4: fusion into 3 scores** — code only
   Fatigue, quality, compensation. Network exists, untrained on real data.
6. **Phase 5: predict compensation in next reps** — not done
   Needs real rep sequences first.
7. **Phase 6: personal fatigue signature** — not done
   Per-session baseline coded; per-person model needs 3+ people.
8. **Action recommendation (5 actions)** — placeholder
   Continue, correct, reduce, rest, see physio. Thresholds invented, no physio review.
9. **Micro-Doppler / spectrogram features** — blocked
   Proposal assumed raw I/Q. Plan B sends points and heatmaps only.
10. **Force sensor / load cell ground truth** — optional
    Listed as possible ground truth; none acquired.

B. Problems in current work

Ranked by how many other tasks they block

1. **Sensors never recorded together on a body** — blocker
   Everything works alone only. Blocks A1–A3, A5–A7.
2. **No phase or raw radar output** — blocker
   Ethernet unverified; firmware change needs approval. Submitted DCA1000 design was replaced by Plan B.
3. **Raspberry Pi never run** — blocker
   SD not flashed. Pi package install needs your OK first.
4. **Model trained on fake data only**
   Point layers hurt on fake data. Frames per rep (128) crops 58% of fake reps.
5. **IMU wiring fails on the arm**
   179 I^2C errors, lost 36 of 72 s. Fix or drop.
6. **Sensor time-sync unproven on real data**
   3-shrug alignment tested on bench and fake data only.
7. **Radar gaps and unverified left/right**
   Sequence gaps on long Windows runs; angle orientation untested.
8. **sEMG thresholds and labels are guesses**
   From fake data and placeholders. Need first real recording and a physio.
9. **Ethics and consent undocumented**
   Healthy-volunteer approval; model needs 3+ people.
10. **Presentation, brochure, housekeeping**
    Deck/script out of date, brochure not started, unpushed files, 2 unreviewed branches, poster wording.

Honesty note: the proposal's fatigue and risk percentages are illustrative, not results. Until step 1 is done, present the system as a working acquisition and dashboard platform, not as validated fatigue or compensation detection.

Sources: CLAUDE/CONTEXT.md, CLAUDE/PROJECT.md, message-2.txt (7-phase roadmap), reports in CLAUDE/output/. Phase status is judged from the repo, not from the Thai proposal text directly.